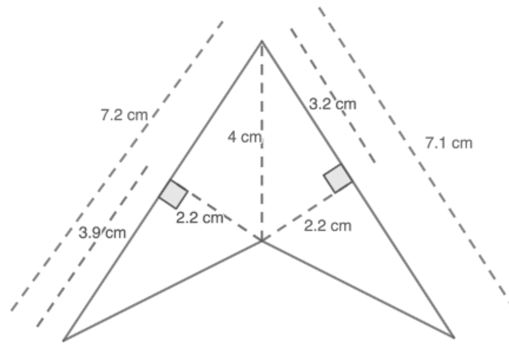
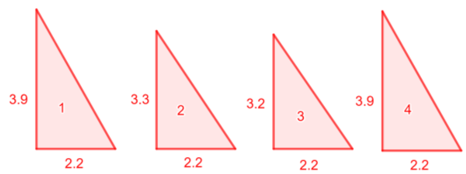
Grade 6 Accelerated
Unit 7
Lesson 3 Homework

Name_Answer Key_________
Date_____________________
Period____________________

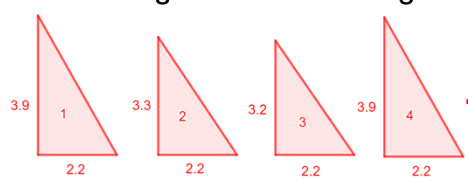
A quadrilateral with measurements in centimeters (cm) is shown for questions 1–3.



1. Decompose the quadrilateral into triangles. Sketch the decomposed shapes here.



2. Using the measurements given in the original image, label the base and height of each triangle.



$7.1 - 3.2 = 3.9$

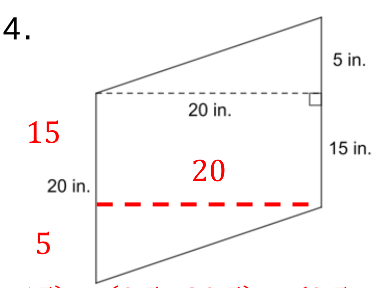
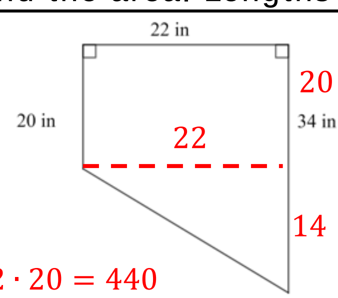
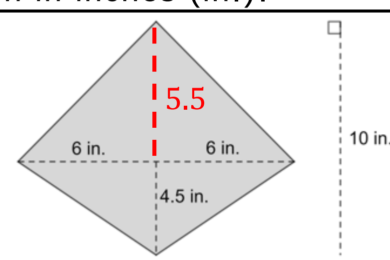
$7.2 - 3.9 = 3.3$

3. Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles. Include units.

$\triangle 1 = 0.5 \cdot 3.9 \cdot 2.2 = 4.29$
$\triangle 2 = 0.5 \cdot 3.3 \cdot 2.2 = 3.63$
$\triangle 3 = 0.5 \cdot 2.2 \cdot 3.2 = 3.52$
$\triangle 4 = 0.5 \cdot 2.2 \cdot 3.9 = 4.29$

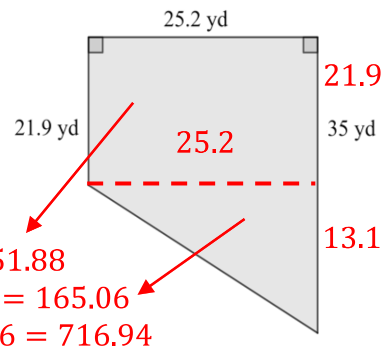
$4.29 + 3.63 + 3.52 + 4.29 = 15.73\ \text{cm}^2$

Decompose each quadrilateral to find the area. Lengths are shown in inches (in.).

4.	5.	6.
 $(20 \cdot 15) + (0.5 \cdot 20.5) + (0.5 \cdot 20.5)$ $300 + 50 + 50$	 $22 \cdot 20 = 440$ $0.5 \cdot 22 \cdot 14 = 154$ $440 + 154 = 594$	 $0.5 \cdot 12 \cdot 5.5 = 33$ $0.5 \cdot 4.5 \cdot 12 = 27$ $33 + 27 = 60$
Area = __400 in^2____	Area = __594 in^2____	Area = __60 in^2____

Decompose each quadrilateral to find the area. Lengths are shown in yards (yd).

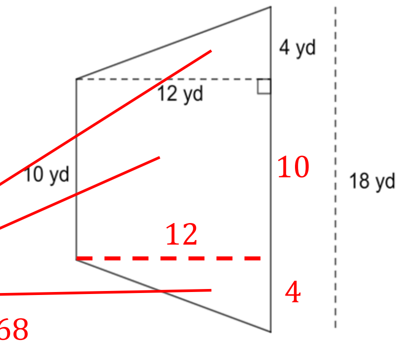
7.



$25.2 \cdot 21.9 = 551.88$
$0.5 \cdot 25.2 \cdot 13.1 = 165.06$
$551.88 + 165.06 = 716.94$

Area = __716.94 yd^2__

8.

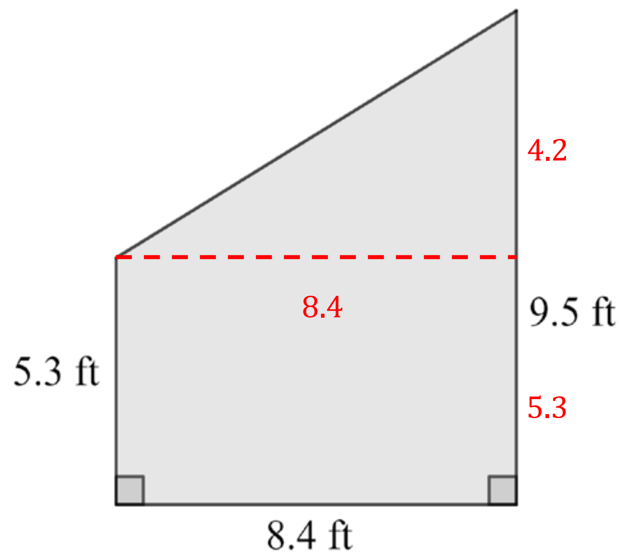


$0.5 \cdot 12 \cdot 4 = 24$
$10 \cdot 12 = 120$
$0.5 \cdot 12 \cdot 4 = 24$
$120 + 24 + 24 = 168$

Area = __168 yd^2__

Sal made a sketch of a wall he needs to paint. The sketch of the wall is shown with measurements in feet for questions 9–10.

9. The home improvement store is offering a sale on their quarts of paint. A quart of paint covers 100 square feet of wall. Will Sal have enough paint if he buys one quart?



Area Rectangle $= 5.3 \cdot 8.4 = 44.52$
Area Triangle $= 0.5 \cdot 8.4 \cdot 4.2 = 17.64$
Total Area $= 44.52 + 17.64 = 62.16$ ft^2

Sal will have enough because the area of the wall is less than 100 ft^2.

10. Determine the amount of paint, in square feet, Sal is short or has left over.

$$\begin{array}{r} 100.00 \\ -\ \ 62.16 \\ \hline 37.84 \end{array}$$

He will have 37.84 ft^2 of paint left over.